

Architecting a Profession

White Paper: Developing the blueprint & roadmap for the Profession of Enterprise Architecture

Abstract | The First Step in the Journey:

Enterprise Architecture is becoming increasingly important for organizations dealing with accelerating levels of change amidst a more complex (and often uncertain) number of variables (covering challenges & opportunities). While thousands of people worldwide are “practicing” as Enterprise Architects, there is no commonly accepted baseline of knowledge nor are there standards and guidelines to ensure consistency of service delivery within the context of a profession. Formalization of a recognized profession of Enterprise Architecture requires the development of the elements of a profession agreed upon by the body of practitioners. In the past, formal professions have gradually evolved over time. In recent years, considerable study has been given to the question of what constitutes a profession. The IEEE Computer Society developed a model of the elements of technical professions that was used as the basis for a roadmap to architect the profession of Enterprise Architecture.

Most recognized professions evolve to their current state over many years (decades and longer in some cases). The challenge that we are tackling in this white paper is to present a framework to accelerate the cycle time for building and achieving a steady state for a new profession (Enterprise Architecture in this case) to address the growing demand for reliable and consistent delivery of services.

WHY do we need a new profession (for Enterprise Architecture)?

The most critical issues facing enterprise architecture are i) an almost total lack of formal, recognized governance, ii) significant inconsistency in practices, and iii) inability to provide assurance that functionally equivalent results can be delivered by any certified (and registered) professional enterprise architect practitioner. Today, the practice of enterprise architecture is inconsistent, and lacks a commonly agreed upon definition and understanding. Further, the continued fractionalization has hurt the legitimacy of qualified practitioners since anyone can call himself or herself an enterprise architect, as there are no standards from which to measure and assess basic levels of competency. The goal of pursuing a profession now is to address these concerns so that Enterprise Architects can be properly positioned to deliver value to all enterprises that are in the midst of accelerating levels of change in increasingly technical and complex environments. More and more organizations are turning to Enterprise Architecture to help plan, design, and augment the execution of change to ensure they can survive and thrive amidst increasing uncertainty.

The legitimacy and formalization of Enterprise Architecture is not consistent across organizations. This results from a lack of agreed knowledge about Enterprise Architecture. As a result, views of Enterprise

Architecture are typically constrained by political, cultural, and social elements instead of professional status. Enterprise Architecture does not yet have the professional standing found in well-established professions like medicine, law, engineering, and accounting. Rather, as in the early days of other, older professions, there are many grass-roots organizations that have sprung up to support practitioners as they have struggled to reach agreement about Enterprise Architecture practices, artifacts, and benefits to the enterprise.

WHY do we need to pursue a profession now?

The Federation of Enterprise Architecture Professional Organizations (FEAPO) is a worldwide association of professional organizations, which have come together to provide a forum to standardize, professionalize, and otherwise advance the profession of Enterprise Architecture. FEAPO provides a focused hub for the Enterprise Architecture expertise and resources of its member organizations, which include, in addition to the IEEE Computer Society and the Center for the Advancement of the Enterprise Architecture Profession, such organizations as The Association of Business Process Management Professionals, The Business Architecture Guild, the Business Architecture Society, the Canadian Information Processing Society, Data Management International (DAMA), and the International Council on Systems Engineering (INCOSE) .

Prior to the establishment of FEAPO, these fragmented efforts failed to coalesce and establish a clear path to sustained legitimacy and a clear direction in pursuit of professionalism. There is no commonly agreed definition of a well-established and consistent set of Enterprise Architect capabilities or a clear differentiator between Enterprise Architecture and the many flavors of “Architecture” in the enterprise or its information technology component. FEAPO’s member organizations have come together to guide the nascent profession to a self-governed status of maturity by 2017 - moving from organized, to qualified, to self-governed, with increasing maturity in three distinct areas:

1. Standards of practice
2. Professional learning
3. Industry self-governance and concomitant professional autonomy

These areas will be addressed using the model of a profession described later in this paper.

WHAT are the building blocks for a profession?

What is the essential distinction between a “line of work” and a “recognized profession”? This is a question that the IEEE Computer Society has previously addressed in the formation and recognition of Software Engineering as a profession. As a founding member of the Federation of Enterprise Architecture Professional Organizations (FEAPO), the Computer Society is sharing its experience and its model of a profession with FEAPO. Significantly, the Center for the Advancement of the Enterprise Architecture Profession (CAEAP), another founding member of FEAPO, has also done extensive work in defining Enterprise Architecture as a profession. Both organizations have identified similar hallmarks of

a profession including the need for a structured educational program. Fortuitously, during the same timeframe, Penn State University began addressing the need for undergraduate, graduate, and continuing education curricula to educate students and practitioners in the field of Enterprise Architecture.

Some practitioners do not see a need to professionalize Enterprise Architecture as rigorously as medicine, law, engineering, accounting or even software engineering. Others are in favor of professionalization to the fullest extent. This debate is centered on whether to position Enterprise Architecture as a legally recognized profession or as a generally accepted profession.

If Enterprise Architecture is to be recognized as a generally accepted profession, it sets its own standards and governs itself, at a minimum, by the four E's:

1. Education
2. Examination
3. Ethics
4. Experience

If Enterprise Architecture is to be recognized legally as profession, it must address additional concerns, such as government certification or licensing by state authorities.

Interactions among the above topics are more complex than they might seem at first glance. For example, the requirements for professional licensing can have a significant impact on the content of initial education. Certification guidelines can influence the content of professional development materials. A professional society may manage the certification process or develop the code of ethics.

Whether legally recognized or generally accepted, the following common traits are shared by both types of professional status:

1. A body of knowledge founded on well-developed and widely accepted theoretical and practical bases.
2. A system for certifying that individuals possess such knowledge before they enter practice.
3. A progressive system of certifying practitioners as they increase both their knowledge and experience in effective practice.
4. A code of ethics, with a commitment to use specialized knowledge for the public good.
5. A professional society, with provisions for monitoring individual compliance to ethical standards and professional practices.

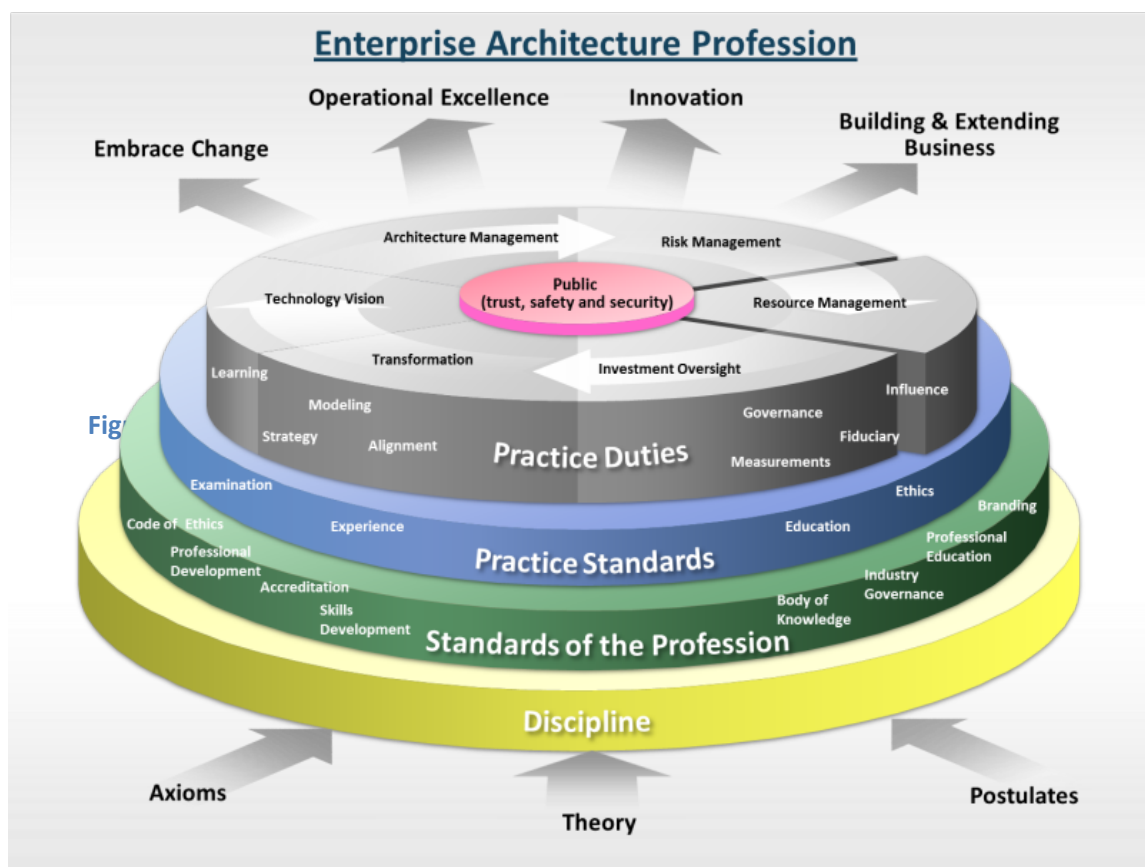
The model that FEAPO has adopted for accelerating the recognition of Enterprise Architecture as a recognized profession includes the elements needed for both: the full model reflects the elements needed for a legally recognized profession; omission of the licensing element is appropriate for a generally accepted profession.

HOW would we build the Profession for Enterprise Architecture?

The profession needs to have a consolidated platform for capturing an Enterprise Architect's portfolio of knowledge, skills, and experience that has been validated using a rigorous qualification process to ensure truth, transparency, and promote trust. In addition, a comprehensive set of services needs to be established to support practitioner and executive career goals including (but not limited to):

- Provides guideposts for practitioners who want to achieve professional status
- Promote individual achievement along the road to becoming professional
- Establishes a key structural element which supports the profession of enterprise architecture

Figure 1: The building blocks for enabling the Enterprise Architecture profession provide a foundation for delivering value along four dimensions: Embracing Change, by implementing the right structures and processes to reduce friction associated with change (across the business planning, operational, and technical areas). Enabling Operational Effectiveness goes beyond efficiency to ensure the business assets (including human ones) are utilized properly and for the right purposes. Effective goes beyond efficient (Lean & Kaizen). Enabling sustainable innovation ensures the right structures are in place



across the organization to manage the multiple portfolios that need to align and work in a synergistic manner. The last dimension for delivering value is Building & Extending the Business (organization), which ties into the other 3 dimensions. The focus is on alignment of strategy, risk, culture, and capabilities (covering business & technology) to create and extend an organization's ability to deliver

sustainable value. The profession is built around a core focus on serving the public. Customers may be the initial consumers of the Enterprise Architect's services, but the professional Enterprise Architect must always ensure the needs of the public are not at the expense of the customer. The Foundational Elements (building blocks) for the profession will be used to create the right skills, capabilities, and knowledge to ensure consistent and ethical delivery of services are provided while measurement, certification, and governance provide the necessary oversight to assure the needs of the public are addressed and standards for the profession are upheld.

WHAT are the Foundational Elements for a Blueprint (Model)?

The combined model (below) of a profession (the basic model developed by the Computer Society and additions from CAEAP) defines the elements that comprise a formal profession. FEAPO has ratified this model as an Organization Technique to guide the sequence and investment of required to "architect" the practice of Enterprise Architecture as a recognized profession. The individual elements are described below in Figure 2 below.

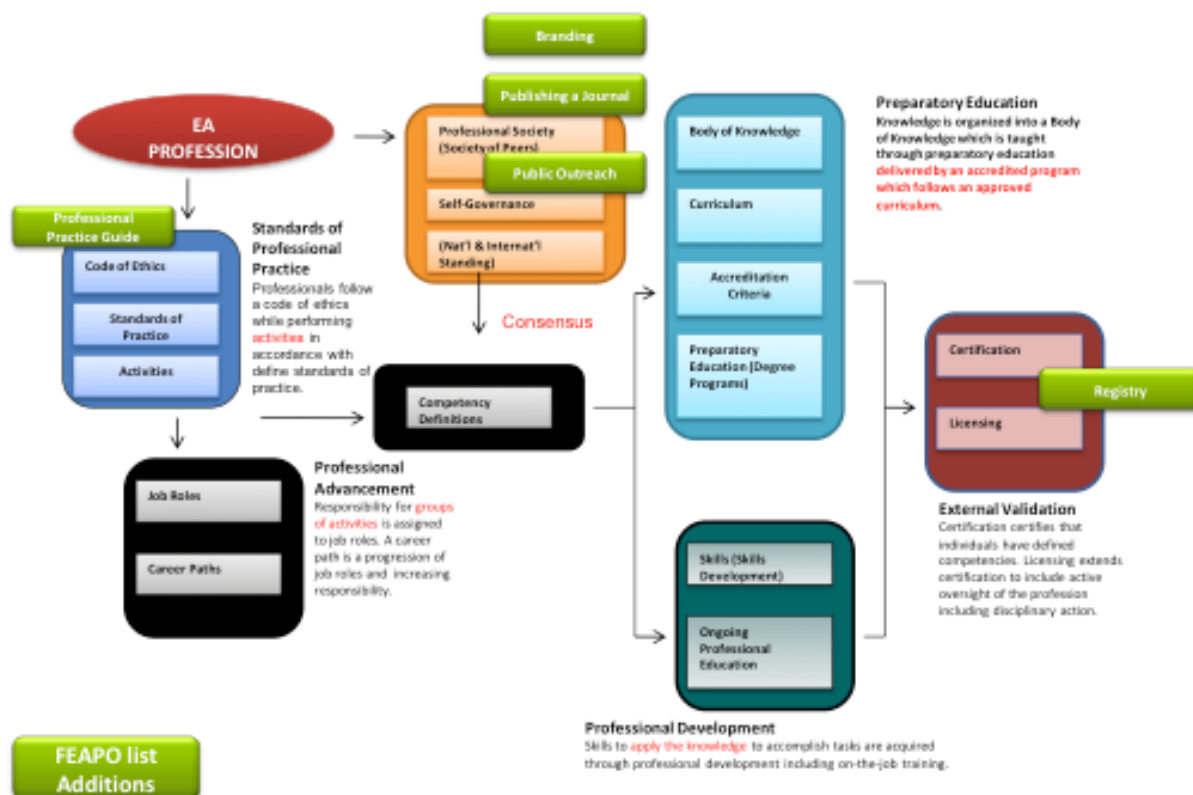


Figure 2: Blended Model of the Enterprise Architecture Profession (IEEE CS + CAEAP Elements)

Professional Societies

- Existence of professional organizations that support the advancement of the profession through such means as branding, publishing refereed papers, such as in a journal, and various forms of public outreach.
- Sufficiency and sustainability of employment in the profession.
- Active research efforts to advance the state of the profession's knowledge

Code of Ethics

- Common, applicable and useful to all Enterprise Architecture sub-disciplines
- Specifies appropriate professional conduct to engage in the profession, including guidance for using specialized knowledge for the common good

Activities

- A known set of activities required in the performance of an Enterprise Architect's work.
- The activity names are commonly understood and can be used in job descriptions.

Standards of practice & Technical Standards

- A Professional Practice Guide
- Application of current best practice in performing activities
- Currency in evolving technologies, methods and techniques
- International standards that specify techniques, methods, procedures and performance norms, consensually agreed by the profession

Competency Model

- Comprehensive competency model in place that describes knowledge, skills and levels of competence
- Ongoing maintenance regime for the competence model

Guide to the Body of Knowledge

- Clearly defined core body of knowledge that establishes the content and boundaries of the discipline
- Specialized bodies of knowledge beyond "core"
- Identification of supporting knowledge areas such as technical management and enterprise operations.
- Consensual validation of the knowledge.
- A rational and scientific foundation for the knowledge.

Curriculum Model(s)

- Defined by recognized, authoritative bodies
- Provide a common basis for the establishment and improvement of educational programs.

- Aligned with relevant Competency Model(s)

Accreditation Criteria

- Aligned to certification requirements
- An accreditation system that assures the quality and suitability of the preparatory education.

Preparatory Education

- Readily available programs
- Complies with curriculum models and accreditation criteria
- Provides the education and training necessary to be employed in an entry level position in the profession.

Skills

- The ability to influence things; to take what you know and apply it so as to cause the desired effect to occur.

NOTE: A certain amount of knowledge is a prerequisite of skill; you can't be skillful without first being knowledgeable; however, you can easily be knowledgeable without being very skillful.

Professional Development

- Multiple sources of education and training to maintain and advance skills and knowledge
- Increasing experience necessary to keep current and advance in the profession.

Certification

- Standard and recognized certification regime
- Consistent with body of knowledge and competency model
- Mechanisms for competency certification that do not require compliance with education requirements (i.e., that enable recognition of competency acquired by other means)
- Defined means of re-certification/CPD & de-certification
- Includes professionalism aspects
- Includes demonstration of competency through practice (apprenticeship / experience)
- Mechanisms to support specialist competence areas
- Validation, by a community of peers, that an individual possesses the knowledge and competence of a professional.
- A public registry of certified professionals

Licensing

- Implemented by other authoritative bodies

- Supported by clear body of knowledge, certification standards, competency standards, code of ethics and a clear boundary of a licensed discipline
- A public registry of licensed professionals
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Job Roles

- Defined positions in an organization defined in terms of the responsibilities for the performance of specific activities assigned to the position and in terms of the competencies required to perform the activities
- Clear delineation of level of responsibility for the role

Career Paths

- A pre-defined series of job roles where experience in previous job roles is relevant to subsequent job roles.
- Paths for technical advancement in job roles based on competencies

HOW do we “build” the Blueprint (Model)?

The elements discussed above should provide us with the foundation to build out a vibrant and (generally) recognized profession. The next step involves defining a roadmap to build out the elements using a phased approach (see Figure 3 below), aligned with key milestones. A number of concurrent activities have already been started and should fit in nicely with the roadmap below. FEAPO member organizations are working together to accelerate the development of the various elements that are needed to unify and support Enterprise Architecture as a profession. Each of the individual members have developed artifacts that contribute to the development of the elements, but until now have lacked a means to prioritize their efforts to bring the disparate artifacts into a coherent whole. The agreed approach was to develop a roadmap for the professionalization of Enterprise Architecture, as presented here:

Overview of the EA Roadmap

Milestones/KPI's

Inception

Foundation

Education

Acknowledgement

Registration

Profession

Maintenance

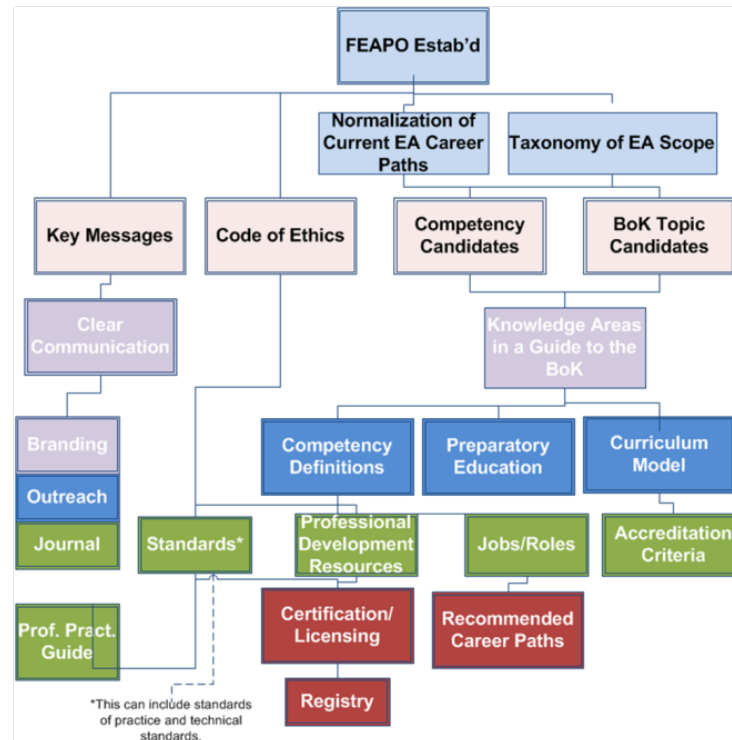


Figure 3: Overview of the Enterprise Architecture Roadmap

Figure 3 represents the major milestones of that roadmap -- a logical order of development for the elements of a profession, focusing on FEAPO's collaborative efforts to promote the professionalism of Enterprise Architecture.

The first step in the roadmap has already been accomplished: FEAPO has been established, and includes the key players from the set of organizations serving Enterprise Architecture practitioners. These member organizations have individually established valuable artifacts and stores of knowledge supporting the practice of Enterprise Architecture. FEAPO's challenge is to build on those in order to provide a single coherent "face" to the profession. Achieving this common face (and voice) necessarily involves collaboration in harmonizing the elements that already exist.

Figure 4 lays out a roadmap for this work. For example, before we can achieve agreement on what belongs in the Guide to the Enterprise Architecture Body of Knowledge (EABOK), we need to agree on the scope (“taxonomy”) of Enterprise Architecture — what’s in and what’s out. In order to agree on the knowledge areas to be described in the Guide, we should have agreement on what skills and competencies a skillful Enterprise Architecture practitioner demonstrates. These can be developed in tandem with (and will feed into) a list of topics for the BoK Guide. The skills/competencies and topics can then be sorted into Knowledge Areas (KAs) to be addressed in the Guide to the BoK. Similarly, by examining existing Enterprise Architecture career paths from corporations and government bodies, we can validate the posited competencies and knowledge areas.

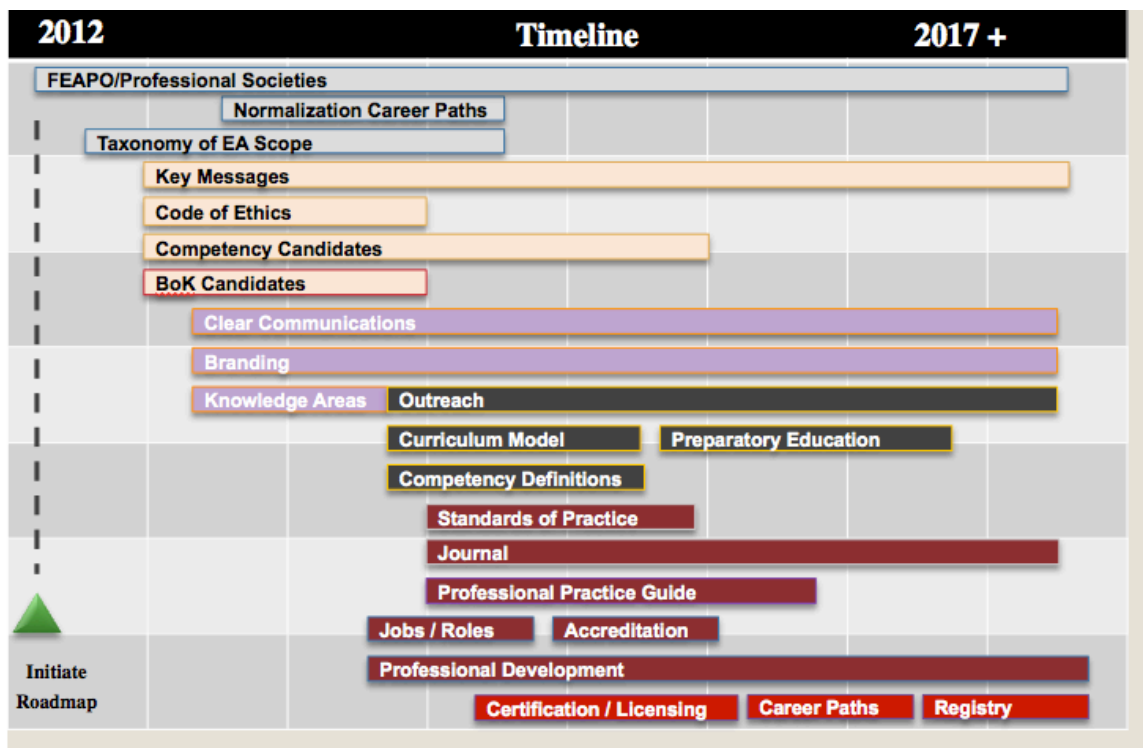


Figure 4: Roadmap & Timeline for Building the Enterprise Architecture Profession

Agreement on the KAs facilitates agreement on how to use the FEAPO Member Organizations’ (FMO) existing knowledge bases and artifacts. Some may be folded into or otherwise support the consensually agreed KAs in the Guide. In other cases, the Guide may cite an FMO’s artifact(s) as a primary reference. In any case, the agreed KAs provide a basis for commonly accepted definitions of requisite competencies and thus provide a basis for developing a Model Curriculum. The Model Curriculum provides a basis for educational institutions to develop their own specific preparatory education courses.

Development of a competency model assembles the requisite competencies increasing levels of competency. This lays the basis for grouping competencies into prototypical job descriptions. The job descriptions can be used for laying out archetypes for possible career paths.

The Road Ahead | The Journey Continues:

The journey towards building the profession of Enterprise Architecture begins like all others with the first step. We believe this paper provides the business case and foundational elements for making that first step.

To attain recognition as a global profession, Enterprise Architecture professionals must have clear ethical and competence standards that accommodate cultural differences in the regulation of professions. Given the ubiquitous role of Enterprise Architecture, a wide range of individuals and organizations has a stake in the profession. Primary stakeholders are professionals and those who employ them. The practice and portability of professional skills are similar across diverse enterprise environments, as well as cultural and geographical boundaries.

In determining to accelerate establishment of Enterprise Architecture as a global profession, FEAPO Member Organizations are working together to:

1. Instantiate the requisite elements of a profession described above
2. Promote the art and science of Enterprise Architecture by uniting likeminded individuals in the major organizations of the Enterprise Architecture ecosystem
3. Identify and establish a body of ethics, principles, and maturity measures that support the Enterprise Architecture profession and its practice.
4. Develop and certify a body of knowledge that properly represents the Enterprise Architecture Profession and its practice.
5. Improve the practice of Enterprise Architecture by providing consensually based artifacts, education and certification
6. Establish collaboration for skills development and professional development between professional societies
7. Establish collaboration avenues for professional education and model curricula developers to achieve accreditation.

We invite enterprise architects and partners with an interest in the journey to join the discussion and participate in the building of this new profession.



“Perhaps no human artifact is more important to the survival of the species than the organization of enterprises. Nearly everything humans do together is done in the context of some kind of enterprise – families, businesses, governments, social groups, educational institutions, military organizations, charities, and so on. Relatively small, static, or simple enterprises in general may not require an explicitly documented enterprise architecture. But most enterprises of consequence are large, dynamic, complicated, and important. Random, organic, component-focused growth and management are no longer viable options for enterprise management. The current global economic situation is strong

evidence of this. Enterprises must be architected and managed by that architecture if they are to remain viable in the Information Age. The Enterprise Architecture profession can make that happen.”¹

¹ “To Be or Not to Be: Recognize Enterprise Architecture as True Profession?,” by Mark Lane, in *The SIM Guide to Enterprise Architecture*, edited by Leon Kappelman, CRC Press, 2009, p. 60.